

Ruilin (Peter) Xie

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Education

University of California, Irvine

Irvine, CA

Master of Software Engineering

Sep 2024 – Dec 2025

Coursework: Data Structures/Algorithms, Distributed Software Architecture, Software Testing & Debugging, Reverse Engineering & Modeling, Concurrent/Network Programming

University of California, Irvine

Bachelor of Arts, Business Economics

Jun 2018 – Aug 2022

Skills

Languages: C# · C++ · Python · Blueprint (UE5) · Java · TypeScript

Engines & Tools: Unity 2022 · Unreal Engine 5 · UMG · Git · Diversion

Disciplines: Gameplay Systems · Runtime Architecture · Data-Driven Design · Technical Design · Balance Tooling

Projects

CrossBolt — Rhythm Game Prototype

Unity 2022, C# | Team (4) | 2025

github.com/MisakaRinOwO/Crossbolt-Code-Samples

- Designed two-layer note schema (NoteType × NoteProperty) encoding gameplay intent into data and prefab routing; defined level-addressing convention (song ID / difficulty / chart.txt) consumed by runtime loader
- Implemented full chart pipeline: custom .txt grammar → NoteLevelConverter → typed timestamp queue → ChartSpawner with type-aware prefab selection and NotesByTimeBuckets for O(1) per-frame hit lookup
- Optimized judgement path on dedicated branch: O(N) active-list scan → O(P) pending-window checks + O(1) hold re-activation lookups
- Built in-engine Beat Map Editor (Unity EditorWindow) with BPM-snap grid; autoplay mode validates charts through the same hit-detection path used at runtime

Void Architect — Modular Ship Combat Prototype

UE5, Blueprint, Python | Solo | 2026

- Implemented tick-driven combat simulation: tracking vs. agility hit probability, per-weapon cooldown, shield→HP overflow with replay event recording per tick
- Built equilateral-triangle placement grid with orientation-aware adjacency rules; data-driven module architecture via UE5 Data Assets with CPU/power budget constraints
- Python log-parsing and run-aggregation scripts for balance regression; Discover→Diagnose→Fix→Verify cycle identified Sniper shield outlier — nerfed values and verified win-rate spread restored across full build matrix
- Per-tick replay UI (timeline slider + HP/Shield curves + event log) serves as both player feedback and developer inspection tool

Project Summon — Action RPG Capstone

UE5, Blueprint, C++ | Team (5) | 2025

- Shipped complete gameplay loop (hub → room combat → loot → return) on fixed 10-week schedule; owned room progression, combat, inventory, and multiplayer persistence systems
- Built BP_RoomManager as self-contained room authority: overlap-scoped enemy tracking, O(1) clear check, multi-spawner orchestration (enemy/chest/portal) on clear trigger
- Implemented cross-level player data sync via GameInstance keyed by controller identity; diagnosed UE5 controller reassignment behavior before implementing persistence path
- C++ integration: exposed enemy HP and TakeDamage (UPROPERTY/UFUNCTION) on teammate-authored code, merging into existing hit-detection path for consistent damage resolution

Godot Engine — Open Source C++ Contributions

C++, Godot 4 | 2025

github.com/godotengine/godot/pulls?q=is%3Apr+author%3AMisakaRinOwO

- PR #107101 (CSGShapes3D): fixed broken hover-feedback by replacing faulty color-mask path with deterministic opacity model; PR #107296 (3D Ruler): added axis-projected helper lines and contextual labels for 3D spatial readability
- Both PRs passed CI and received maintainer review; two earlier test-coverage PRs (Sprite2D, AnimationBlendTree) also merged into master